Yiyang Wei

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Education

Zhejiang University Hangzhou, China

B.Eng in Information Engineering *Sep. 2020 – July 2024*

- Major GPA(Third Year): 3.72/4
- Selected Courses: Calculus(A+), Linear Algebra(A), C Programming(A+), Fundamental Data Structures(A), Operating System(A+)

Selected Work Experience

Yale University Sep 2025 – Now

Research Collaboration *Remote*

- **First-Authored Survey(ACL Under Review):** A comprehensive survey on Reasoning-Intensive Retrieval (RIR), covering benchmarks, methods across the retrieval pipeline, applications, and open challenges; **(ACL 2026 Meta-review score: 4/5)**.
- **Synthetic Training Data for Retriever:** Engineered a multi-agent synthetic data pipeline for training reasoning-intensive retrievers, achieving a 12% improvement in recall performance on the BRIGHT benchmark.
- **Multimodal Query Rewriting:** Architecting an RLVR-based (Reinforcement Learning with Verifiable Rewards) multimodal rewriting model to resolve cross-modal context loss and bridge existing gaps in RIR pipelines.

Alibaba July 2024 – July 2025

AI Engineer Full-Time *Hangzhou, China*

- **DingTalk AI Image Search:** Fully engineered the foundational online search and offline index construction pipelines. Subsequently optimized the system by deploying a closeness-aware online ranking module and a fine-tuned CLIP offline indexing pipeline, resulting in an 18.9% increase in CTR, a 5% boost in recall, and a 95% reduction in indexing costs.
- **Large-Scale RAG-Based AI Search System:** Led the design of intent recognition and parameter extraction modules, and built **Text-to-SQL**-based user behavior(including send, receive, etc.) retrieval and indexing pipelines, supporting 2M+ users and 100k daily active users, enabling personalized and context-aware search.
- **RAG Evaluation System:** Developed an end-to-end RAG evaluation pipeline that quantified hallucination risk, faithfulness,response-level relevance for (query, retrieval, answer) core triplets in AI Search System.

Zhejiang University Nov 2023 – May 2024

Research Intern *Hangzhou, China*

- Developed a dual-layer federated/pipelined learning framework for LLM fine-tuning on 7+ edge devices, reducing accuracy drop to 0.63% (vs. 1.44% in traditional federated learning, a 56% improvement).
- Upgraded communication to gRPC, 2x faster on edge transmission and 40% bandwidth reduction vs HTTP/1.1.

Trip.com April 2023 – July 2023

AI Engineer Intern *Shanghai, China*

- Developed and deployed a GPT service platform handling over 100K daily requests using Spring Boot, SOA, Elastic Search, CAT, and Hystrix for API integration, logging, monitoring, alarming, rate limiting, and load balancing.
- Built an Email-GPT assistant service with Retrieval Augmented Generation (RAG), LangChain, Gradio, and PostgreSQL to parse, translate, and facilitate interaction with international emails, providing writing suggestions

Skills

Programming:	Python, C/C++, Java, Go, SQL
IR Techniques:	Retrieval-Augmented Generation (RAG), Dense Retrieval (BGE-embedding, HNSW, K-means), Sparse Retrieval (BM25, Inverted Index), Multi-modal Search (CLIP, Vision Transformer (ViT))
Framework:	Pytorch, LangChain, Flask, Springboot, Gin, Gorm, gRPC
Middlewares:	Message queue, Database, Redis, Elastic Search, RPC, Sentinel, Hystrix
Developer Tools:	Git, Maven, Docker, PostMan, VS Code, PyCharm, IntelliJ, Clion, Goland, Vim, LateX
Languages:	Chinese Mandarin (Native), English (Fluent, TOEFL iBT 101)